FEDERAL SPECIFICATION

STAND, BED SERVICE, (ALUMINUM)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers bed service stands fabricated of aluminum and designed with folding leg supports. The top shall be provided with a ratchet type elevating mechanism to hold it at various selected angles.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND DRAWINGS

2.1 **Specifications and standards.**—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

QQ-A-318—Aluminum Alloy 52S; Plate and Sheet.

QQ-A-561—Aluminum Alloy 2S; Plate and Sheet.

QQ-S-766 — Steel, Corrosion-Resisting, Plates, Sheets, Strips, and Structural Shapes.

LLL-B-631—Boxes; Fiber, Corrugated (for Domestic Shipment).

LLL-B-636—Boxes, Fiber, Solid (for Domestic Shipment).

PPP-B-585—Boxes; Wood, Wirebound.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

Federal Standard:

Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-B-10377—Box, Wood, Cleated, Veneer, Paper Overlaid.

MIL-L-10547 — Liners, Case, Waterproof.

MIL-A-18593 — Aluminum Alloy Bars, Rods, and Structural and Special Shaped Sections — Extruded, 6063 (63S).

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of Military Specifications, Standards, and Drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 **Drawings.**—The following Armed Services Medical Procurement Agency drawings, of the issue in effect on date of invitation for bids, forms a part of this specification:

No. 1056—Insignia.

No. 1721—Stand, Bed Service, Aluminum.

(Reduced copy of drawing No. 1721 is included herein as figure 1. Copy of original size drawing will be furnished manufacturers upon request by the procuring activity or as directed by that activity.)

3. REQUIREMENTS

3.1 **Materials.**—Except as specified herein, materials shall conform to the applicable portions of the specifications and drawings listed in section 2, and shall be capable of meeting the requirements shown therein. When materials to be used are not specifically designated, those used shall be of the highest commercial quality suitable for the purpose intended.

3.1.1 *Aluminum alloy.*

3.1.1.1 *Frame.*—The frame and leg supports shall be made of 63 S–T5 aluminum alloy conforming to Military Specification MIL–A–18593.

3.1.1.2 *Top.*—The top shall be made of sheet aluminum alloy conforming to Federal Specification QQ–A–561, condition ½H.

3.1.1.3 *Piano hinge.*—Piano hinge shall be aluminum alloy 52S conforming to Federal Specification QQ–A–318 for condition ½ hard.

3.1.1.4 *Rivets.*—All rivets, except the rivet for pivoting the pawl, shall be of A17S–T4 aluminum alloy.

3.1.2 *Corrosion-resisting steel.*

3.1.2.1 *Retention springs.*—The retention springs shall be made of corrosion-resisting steel conforming to Federal Specification QQ–S–766 for class 4, finish 2, and with a suitable spring temper.

3.1.2.2 *Other corrosion-resisting steel.* — The leg support hinges, the piano hinge pin, and all components of the safety latch and top adjusting mechanism, including the rivet for pivoting the pawl, shall conform to Federal Specification QQ–S–766 for any suitable class or condition, except class 3, condition a.

3.1.3 *Bumpers.*—Bumpers shall be made of an appropriate, soft rubber composition, and shall be securely mounted.

3.2 **Construction.**—Stand shall consist of a frame, two collapsible leg supports, four leg support retention springs, two leg support safety latches, and an adjustable top. The stand shall be constructed in conformance with figure 1, and as hereinafter specified. In use, the stand shall be capable of being easily lifted and conveniently carried about as a handled stand and of serving as a bed stand support above the patient's lap while in a sitting posture. The adjustable top shall permit the stand to be used as a reading stand, for letter writing, and for recreational activities for bed patients.

3.2.1 *Riveting.* — Except when otherwise specified, the stand shall be an assembly of riveted members. Either countersunk or round headed rivets may be used depending on service requirement and appearance. The riveting shall be durable and sound in every detail, and the component members of the stand shall be carefully and properly aligned, prior to riveting into permanent position, so as to form a straight, rigid, and level assembly. All round head rivets shall be upset symmetrically and smoothly, and all countersunk rivets ground flush with the adjacent surfaces. Rivet holes shall be drilled or punched, and reamed to a diameter equal to 102 to 103 percent of the rivet diameter. Rivets for attaching the piano hinge to the frame, and for attaching the safety latch guides to the top braces of the leg supports, shall be ⅛ inch in diameter. The rivets for securing the retention springs shall be ¼ inch in diameter, and all other aluminum alloy rivets shall be $\frac{3}{16}$ inch in diameter.

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3.2.2 *Assembly.*—The stand shall be assembled of component members as hereinafter specified.

3.2.2.1 *Frame.*—The frame shall be "H" shaped as shown on figure 1 and shall serve as a support for the top and the collapsible legs. The legs of the "H" shall be 2- by ¼-inch aluminum alloy flats (see 3.1.1.1) to which is butt welded a cross aluminum alloy flat measuring 2 by ¼ inches. The ends of the legs shall have an accurately finished square cut edge to fit flush against the top braces of the leg supports when the supports are extended, and to limit the hinging to 90 degrees.

3.2.2.2 *Leg supports.* — Each leg support shall consist of a ¾ by ¼ inch aluminum alloy flat (see 3.1.1.1) shaped as shown on figure 1, and braced by 1½ by ¼ inch aluminum alloy top brace welded between the uprights in a vertical position with fillets ground to a smooth curvature to merge with the uprights. To each cross brace, in the position shown on figure 1, there shall be welded a handle formed of aluminum alloy (see 3.1.1.1) ⅜-inch square rod. The handles shall have an inside width of 3½ inches and a hand clearance of 1⅞ inches between the top brace and the handle grip, with the ends of the grip section bent at a comfortable and convenient radius. The leg supports shall be joined at the top brace to the legs of the "H" frame by means of fast pin 1½-inch wide corrosion-resisting steel hinges not less than 0.070-inch thick (see 3.1.2.2). The hinges shall be secured by means of countersunk rivets as shown on figure 1. The leg support top braces and the leg support hinges shall be notched to accommodate the retention springs.

3.2.2.3 *Retention springs.*—The four retention springs shall serve to retain the leg supports in the collapsed position, and in combination with the safety latch, retain them in the extended position as well. Springs shall be ¾ by ⅛ inch corrosion-resisting steel (see 3.1.2.1). One end of each retention spring shall be riveted to a leg of the "H" frame with the outer end extending through the horizontal notch in the leg support top brace and bottom edge of hinge butt and then toed upright in a sharp bend, face to face against the vertical side surface of the brace. The springs shall be of such resiliency that the leg supports can be brought to, extended, or released to collapsed position manually with little effort, but with sufficient residual tension to seat firmly within the slot of the cross braces when the leg supports are extended.

3.2.2.4 *Safety latches.*—The safety latches shall securely and automatically lock each leg support in the extended position to prevent accidental collapse of either leg support of the standard. Each leg support safety latch shall consist of two spring loaded latch slides which can be displaced horizontally by means of a finger operated release button. Each slide shall be slotted and retained by two guides which are riveted to the leg support top brace. When each leg support is extended, a notch in each spring loaded safety latch slide shall engage a mating notch in its corresponding retention spring to securely lock the retention spring and hence the leg support in position. To release the safety latch, the release button shall be depressed, moving the latch slides outward horizontally, disengaging the slides from the retention springs and permitting the leg supports to be collapsed. All components of the safety latch shall be constructed of corrosion-resisting steel (see 3.1.2.2) and shall be styled as shown on figure 1.

3.2.2.5 *Adjustable top.*—The top shall be made of a sheet of 0.091-inch-thick aluminum alloy (see 3.1.1.2) and shall be formed with upright flanges on the front and rear sides. The top shall be pivoted to one leg of the "H" frame by means of a continuous aluminum alloy piano hinge (see 3.1.1.3) fitted with a corrosion-resisting steel pin (see 3.1.2.2). The piano hinge shall be spot welded to the top and riveted to the leg of the frame with no less than seven rivets. The other leg of the frame shall be provided with suitable

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silencers in the form of rubber bumpers (see 3.1.5) at the top and bottom surfaces to eliminate the noise of metal to metal contact when the top is closed, or when the leg supports are collapsed. A knurled cam lock shall be provided to securely lock the top at the horizontal position when adjustment with the ratchet adusting mechanism is not desired. Cam shall be held to one leg of the "H" frame by means of an aluminum alloy spacer and screw. There shall be an additional rubber bumper, located on the frame near the cam lock, which shall act as a compression spring between the frame and the top. The height of the bumper shall be such as to provide an adequate force between the top and the cam lock to hold the cam in its locked position. A spring lock washer shall be provided under the cam so that it will rotate stiffly, thereby preventing the cam from falling into the open position if pressure should be inadvertently applied to the top.

3.2.2.5.1 *Top adjusting mechanism.*—The top adjusting mechanism shall consist of a ratchet and pawl for holding the top firmly in any one of three approximate angular positions, as follows: 30 degrees, 45 degrees, and 60 degrees from the horizontal. The ratchet shall consist of a 1¼ by 1¼ by ⅛ inch corrosion-resisting steel angle with one leg riveted to the cross brace of the "H" frame, and the longer leg notched as shown in figure 1. The ratchet pawl shall be connected to the table top by means of a corrosion-resisting steel pawl arm which is hinged to the top through an aluminum alloy angle with a ¼ inch corrosion-resisting steel rivet. The pawl shall be provided with a flat surface to prevent rotation relative to the pawl arm when the pawl locking screw is tightened. The pawl locking screw shall clamp the pawl arm to the ratchet when it is tightened thereby locking the table top at a desired angular position. The design of the ratchet adjusting mechanism shall be in conformance with figure 1.

3.3 Finish.

3.3.1 *Corrosion-resisting steel.*—All corrosion-resisting steel components shall have a mill finish brightened by tumbling process used to remove burs.

3.3.2 *Aluminum alloy.* — Aluminum alloy components shall be given a decorative silvery-white frosted finish by means of a caustic etch. Streaking shall be kept to a minimum and, if necessary, a mild caustic dip shall follow the main caustic dip to eliminate streaking. Etching shall be followed by a cold water rinse and then an acid dip to remove smut deposited by the caustic solution and to neutralize any remaining alkali on the metal surface. The components shall then be rinsed thoroughly in cold water and dried.

3.3.3 *Final finish.*—The completed stand including corrosion-resisting steel components shall be given a coat of aluminum pigmented lacquer or enamel, baked on or force dried. The pigmented coat shall be followed by a finish coat of clear vinyl lacquer, a minimum of 0.8 mil thick. The finish coat shall be full baked at the temperature recommended by the paint manufacturer. The finish shall pass the tests specified in 4.4.1.

3.4 Identification marking.—Each stand shall have a metal nameplate permanently attached, located in the position shown on figure 1, and bearing the following information:

Name or registered trade-mark of manufacturer.

Contract number and date.

A caduceus, centered between the letters "U" and "S," as shown on figure 2.

3.5 Weight.—The stand shall weigh approximately 10 pounds, when completely assembled.

3.6 Welding.—All welding shall be done with appropriate equipment, properly regulated and operated by thoroughly trained and experienced welders highly skilled in the operation of aluminum welding. Fusion welds shall be accomplished by the heliarc, or other approved equal, welding process, using filler

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rod of appropriate composition. The surfaces of parts to be welded shall be properly cleaned and otherwise prepared to assure sound welds of maximum strength. Proper precautions shall be exercised to prevent distortion or buckling in the welded assemblies. All welds shall be ground smooth to remove all traces of weld marks and polished to blend into adjacent surfaces.

3.6.1 *Spot welding.*—Spot welds shall be on not greater than 2-inch centers and shall be accomplished so as to insure sound, well fused welds of maximum strength. All spot welds shall be ground smooth, and when necessary, filled to eliminate unsightly extrusions and indentations.

3.7 **Workmanship.**—Workmanship shall be first class in every respect. All exposed corners shall be smoothly rounded throughout and all sharp edges eliminated in order to prevent injury to personnel or damage to clothing. The stands shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—Unless otherwise specified, samples shall be submitted in accordance with instructions furnished by the procuring activity (see 6.2). In addition, sampling procedures shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

4.2 **Inspection.**—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery as specified (see 6.2). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.3 **Classification of defects.**—The classification of defects found during examination shall be in accordance with the list below. The Government reserves the right to require examination for any defect not included herein and to classify such defects in accordance with the definitions contained in Military Standard MIL–STD–105. The sample unit for these examinations shall be one finished bed service stand.

Categories and defects
AQL 2.5 percent

Major

1. Leg supports do not open to 90° when fully extended (3.2.2.1).
2. Leg brace not securely welded to cross brace (3.2.2.2).
3. Handle not securely attached to cross brace (3.2.2.2).
4. Handle incorrectly formed substantially interfering with portability (3.2.2.2).
5. Retention springs (in combination with safety latch) do not properly retain the leg supports in extended position; or both collapsed or extended positions (3.2.2.3).
6. Safety latch does not operate properly (3.2.2.4).
7. Top not adjustable to 30°, 45°, or 60° angles from the horizontal (3.2.2.5.1).
8. Pawl locking screw does not securely lock table top at specified angular position (3.2.2.5.1).
9. Poor welds seriously affecting utility of item (3.6.1), (3.7).
10. Exposed sharp corners not rounded, or exposed sharp edges not broken so that individuals using or handling the item may be exposed to injury, or clothing may be damaged (3.7).

Minor

101. Rivets not properly driven (3.2.1).
102. Rivets not of specified size (3.2.1).
103. Leg supports do not fit flush against top braces (3.2.2.1).

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104. Handle incorrectly formed not interfering with portability (3.2.2.2).
105. Retention springs do not properly retain leg supports in collapsed position (3.2.2.3) or permit legs to operate easily.
106. The top does not pivot properly (3.2.2.5).
107. Rubber bumpers missing or incorrectly located (3.2.2.5).
108. Rubber bumpers not suitable (3.2.2.5).
109. Knurled cam lock does not adequately secure top in horizontal position (3.2.2.5).
110. Cam lock works too loosely (3.2.2.5).
111. Finish not smooth and uniform over entire surface (3.3).
112. Identification markings incomplete or incorrect, poorly applied, not properly located (3.4).
113. Poor welds not seriously affecting utility of item (3.6.1), (3.7).
114. Welds not properly finished (3.6.1).

4.4 Tests.—Tests shall be conducted in accordance with the applicable specifications, as necessary, to determine compliance with the specification.

4.4.1 *Tests on finish.*—The manufacturer shall supply ten 4- by 6-inch aluminum alloy test panels finished in exactly the same manner as the stand. These test panels shall be subjected to the following tests:

4.4.1.1 *Humidity test.*—The finish shall not blister after being subjected to 100 percent relative humidity at 100° F. for a period of 48 hours.

4.4.1.2 *Salt spray test.*—The panel shall be prepared for test by scribing each panel along two diagonals for a distance of 5 inches, using a razor blade. The scribe marks shall penetrate through the coating to the base metal. The panel shall be exposed to a salt spray test given in accordance with the applicable requirements of Federal Test Method Standard 151 for 48 hours and shall show no evidence of corrosion or peeling of the coating at points other than within ⅛ inch distance of the scribe marks.

4.4.1.3 *Chemical spot tests.*—Finish shall show no deterioration when subjected to the following reagents at room temperature, confined under a watch glass for the specified time:

Percent	*Reagent*
20	Sodium hydroxide solution---24 hours.
50	Sulfuric acid ----------24 hours.
95	Ethyl alcohol ----------30 minutes.

4.4.1.4 *Water immersion test.* — Finish shall show no blistering after panel is immersed in water for 24 hours at room temperature.

4.4.1.5 *Grease resistance test.*—The finish shall not permanently soften when smeared with equal parts c.p. oleic acid and lard oil, and stored at room temperature for a period of 5 days.

4.4.1.6 *Appearance test.*—Finish shall be free of deep orangepeel, seed, foreign matter, and shall have sufficient gloss to give the desired eye appeal.

5. PREPARATION FOR DELIVERY

(For definitions and levels, see 6.1).

5.1 Packaging.

5.1.1 *Unit of issue.*—One stand, as specified, constitutes one unit.

5.1.2 *Quantities.*—Packaging and packing quantities shall be as specified in table I.

TABLE I.—*Packaging and packing quantities*

Unit pack	Intermediate pack	Shipping container
1 unit	Not required	6 units

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is

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less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*—Each unit shall be packaged in a solid fiberboard or double-faced corrugated fiberboard carton, of appropriate size and design, constructed in accordance with Federal Specification LLL–B–631 or LLL–B–636. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used for the purpose, which will afford the degree of protection required for shipment and use of the end product for its intended purpose, within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—The number of units specified in table I shall be overpacked in an exterior container designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621; or MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—The number of units specified in table I shall be overpacked in an exterior container conforming to Federal Specification PPP–B–601, table I, using class 1 plywood; or an exterior container designed for type 2 load and constructed in accordance with Federal Specification PPP–B–621; PPP–B–585, style 3, for class 2 use; or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Unit carton.*—Each unit carton shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2 *Shipping container.*—Each shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color marking and symbol are not required on containers for level A packs when addressed to domestic establishments, for the Medical Services of the Armed Forces.

6. NOTES

6.1 Federal Standard 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification covers the following item appearing in the Armed Services Medical Stock List:

6530–708–0000 Stand, Bed Service, Aluminum.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.5 **Transportation description.** — Transportation description applicable to item is:

Stands, not otherwise indexed by name, metal, (Indicate set-up or knocked-down).

Carload minimum weights;
In cars 40′–9″ in length, inside measurements 10,000; 12,000; 15,000; 18,000; 22,000; and 30,000 pounds.
In cars exceeding 40′–9″ in length, inside measurements, 16,000; 18,000; 21,000; 24,000; 30,000 and 40,000 pounds.

Truckload minimum weight 12,000 pounds.

6.6 **Superseded specification.**—This specification incorporates the requirements of Military Specification MIL–T–16669(BuMed) dated October 1, 1951.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

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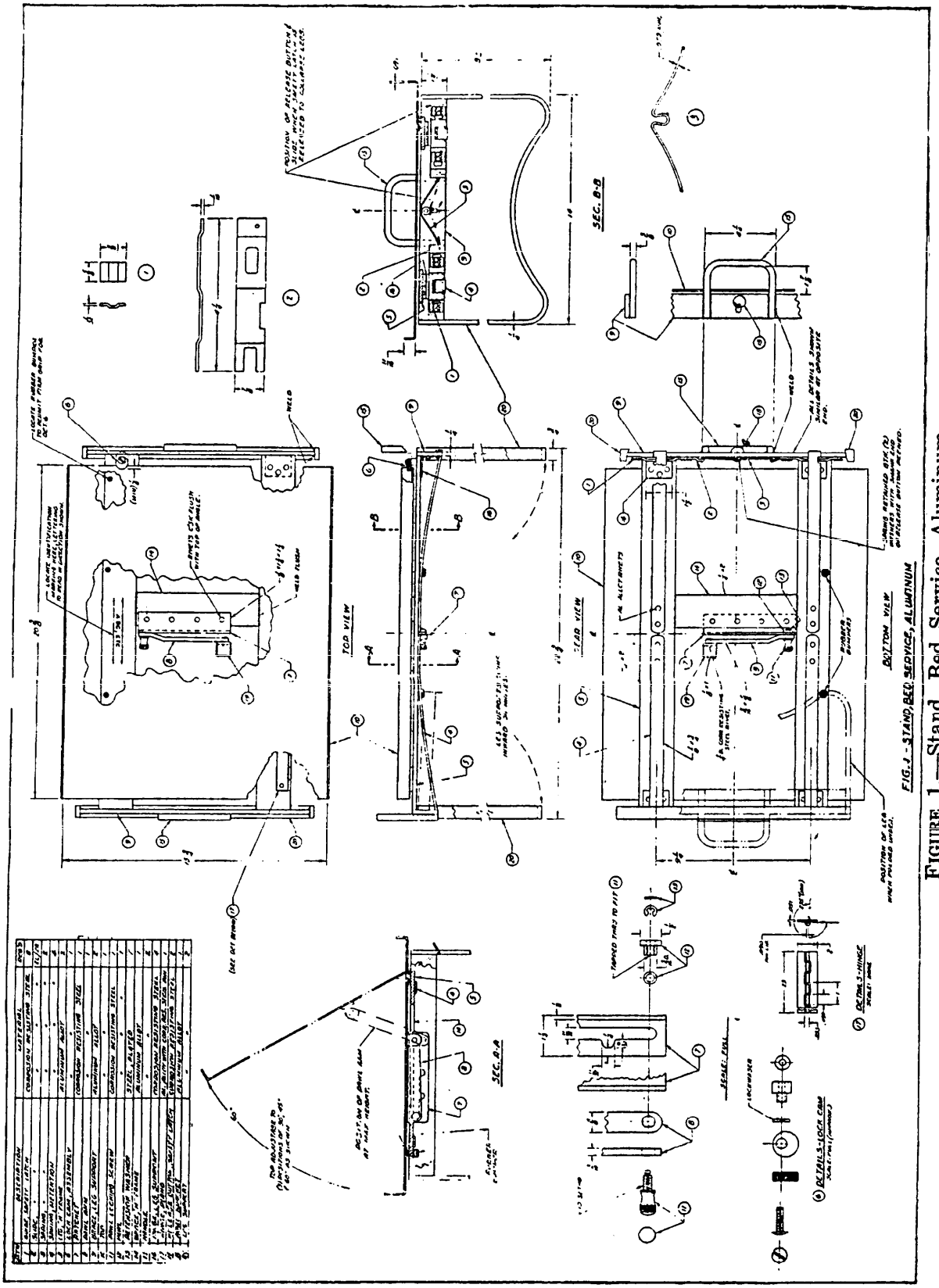


FIGURE 1.—Stand, Bed Service, Aluminum.

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FIGURE 2.—Insignia.

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